12/02/24 AN

Hall Ticket Number:

MR-21

Code: B414CS4S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, Jan/Feb-2024

Java Programming

(Common to CSE & IT)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Explain the usage of 'this' keyword.	2	2	1	1,2
1. b)	Discuss about binary, unary, and ternary operators.	2	2	1	1,2
1. c)	Explain about the importance of extends and implement keywords.	2	2	2	1,2
1. d)	How to create and use a package in Java program?	2	1	2	1,2
1. e)	Explain Enumerations in Java.	2	2	3	1,2
1. f)	Define inter-thread communication?	2	1	4	1,2
1. g)	Explain about Scanner with an example.	2	2	3	1,2
1. h)	Define ArrayList with syntax.	2	1	3	1,2
1. i)	Give a note on layouts in AWT.	2	1	5	1,2
1. j)	Explain in brief about Flow Layout.	2	2	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	What is Polymorphism. Explain runtime polymorphism in detail.	5	2	1	1,2
2. b)	Explain about constructor overloading.	5	2	1	1,2
	OR				
2. c)	Give a brief note on operators in java.	5	1	1	1,2
2. d)	Explain various Access specifiers supported by Java with an example.	5	3	1	1,2,3
3. a)	What is the importance of Exception Handling in Java? Define and distinguish between checked and unchecked exceptions.	5	4	2	1,2
3. b)	Explain Packages and Interfaces with a suitable program.	5	2	2	1,2,3
	OR				
3. c)	Write a java program which illustrates try, catch, throw, throws and finally keywords.	5	3	2	1,2,3
3. d)	Write a Java program to find the area and perimeter of square and circle using interface.	5	3	2	1,2,3
4. a)	Differentiate between thread and process? How is a thread created? Discuss.	5	4	4	1,2
4. b)	What is a stream? What is the difference between byte streams and character streams? How are they used to capture input from the user?	5	4	3	1,2
	OR				
4. c)	How to set priorities for threads? Build an example.	4	3	4	1,2,3
4. d)	Explain in detail about different stream classes.	6	2	3	1,2
5. a)	What is difference between ArrayList and LinkedList in collection framework? Explain.	5	4	3	1,2
5. b)	What are common algorithms implemented in Collections Framework?	5	1	3	1,2
	OR				
5. c)	What is the difference between Iterator and ListIterator?	4	4	3	1,2
5. d)	Explain how HashMap works in Java with an example.	6	2	3	1,2,3
6. a)	Explain the concept of Applet-to-Applet communication with suitable program.	5	3	5	1,2,3
6. b)	What is the significance of Layout managers? Discuss briefly various layout managers.	5	2	5	1,2
	OR				
6. c)	Write about different event classes.	5	2	5	1,2
6. d)	What are the limitations of AWT?	5	2	5	1,2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome